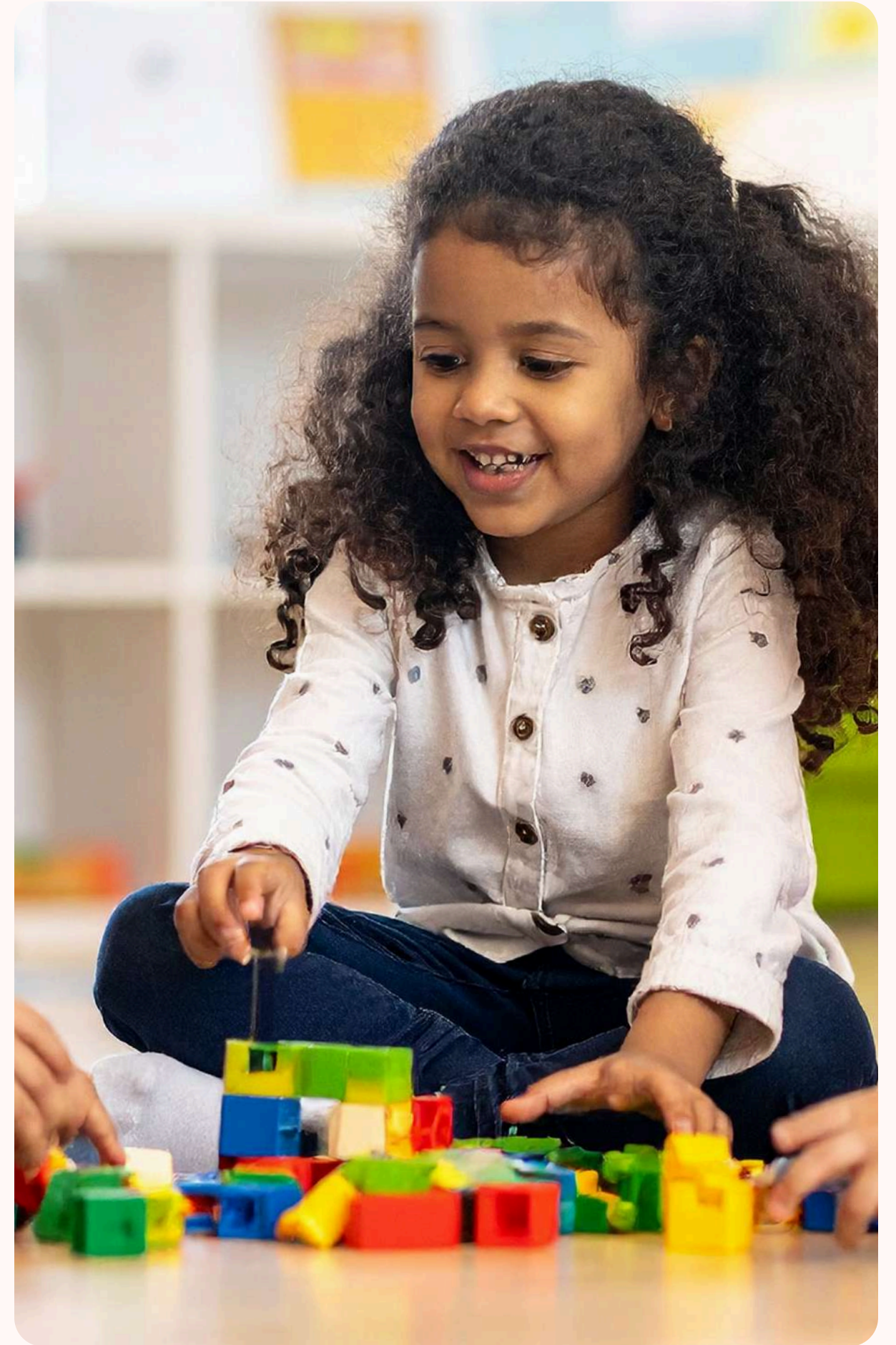


Maths: Statistics



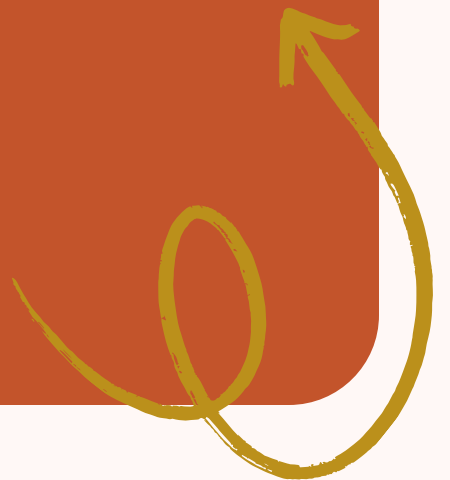
Introduction to Statistics

Statistics is a powerful tool that helps us **understand data** and make informed decisions. By learning about data, we gain insights that can **shape our choices** in everyday life.



Understanding Data Distribution

How data can be spread out





What is Normal Distribution?

Understanding the bell curve shape

A **normal distribution** is a special type of data distribution that appears as a bell curve. It represents how many things in life, like height and test scores, are distributed.

Properties of Normal Distribution

- **Symmetrical shape** with equal halves on both sides
- **Mean, median, and mode** all align at the center
- **Tails extend infinitely**, showing extreme values can occur



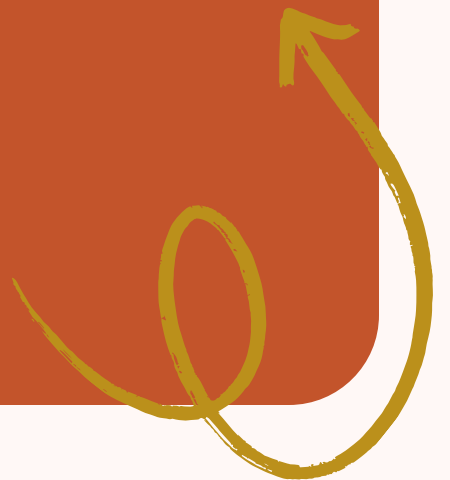


Importance of Normal Distribution

Understanding the **normal distribution** is crucial because it helps us analyze data patterns. It allows us to predict outcomes and make informed decisions based on statistical evidence, enhancing our learning experience.

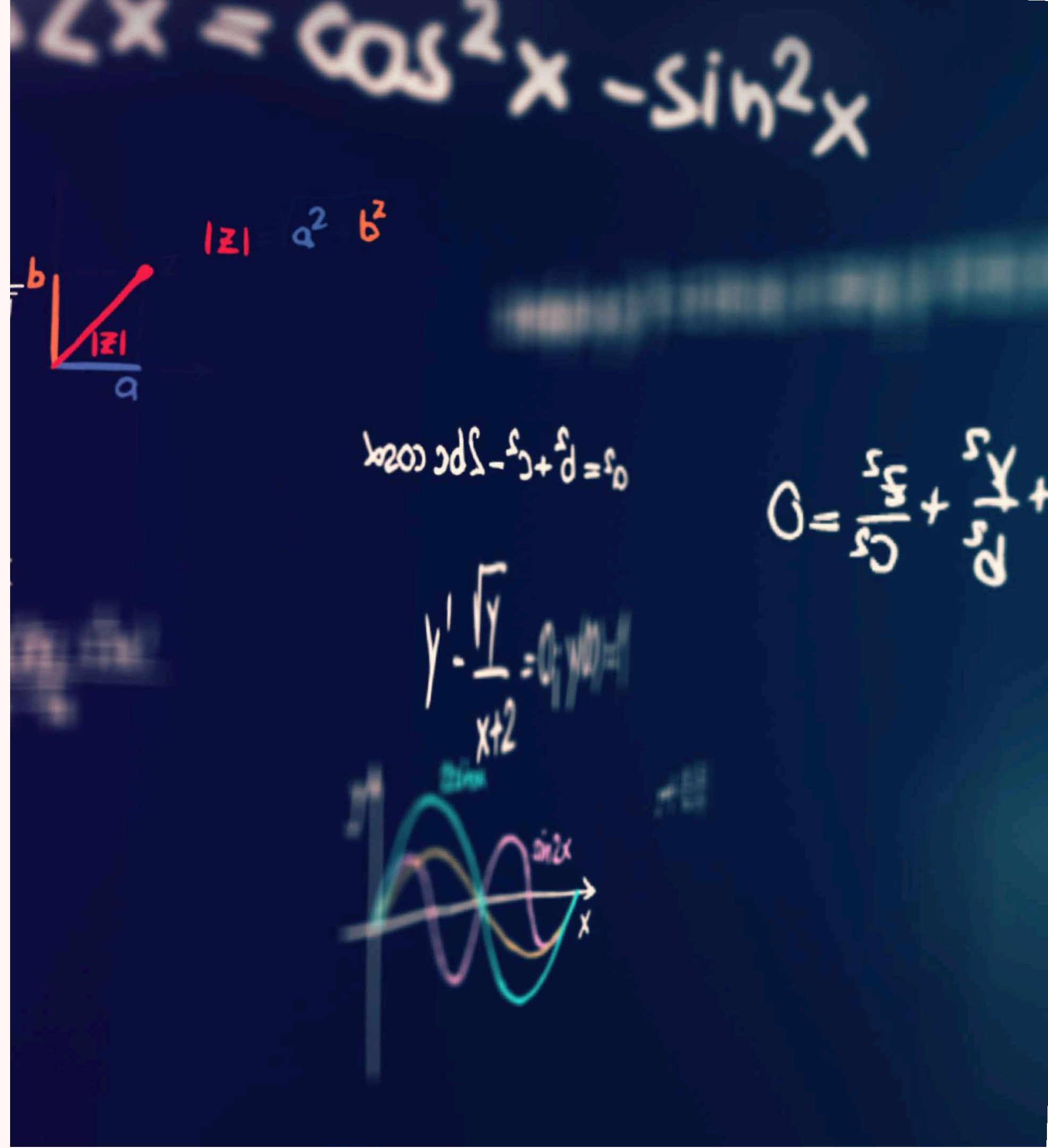
Introducing Z-scores

Understanding how far from average



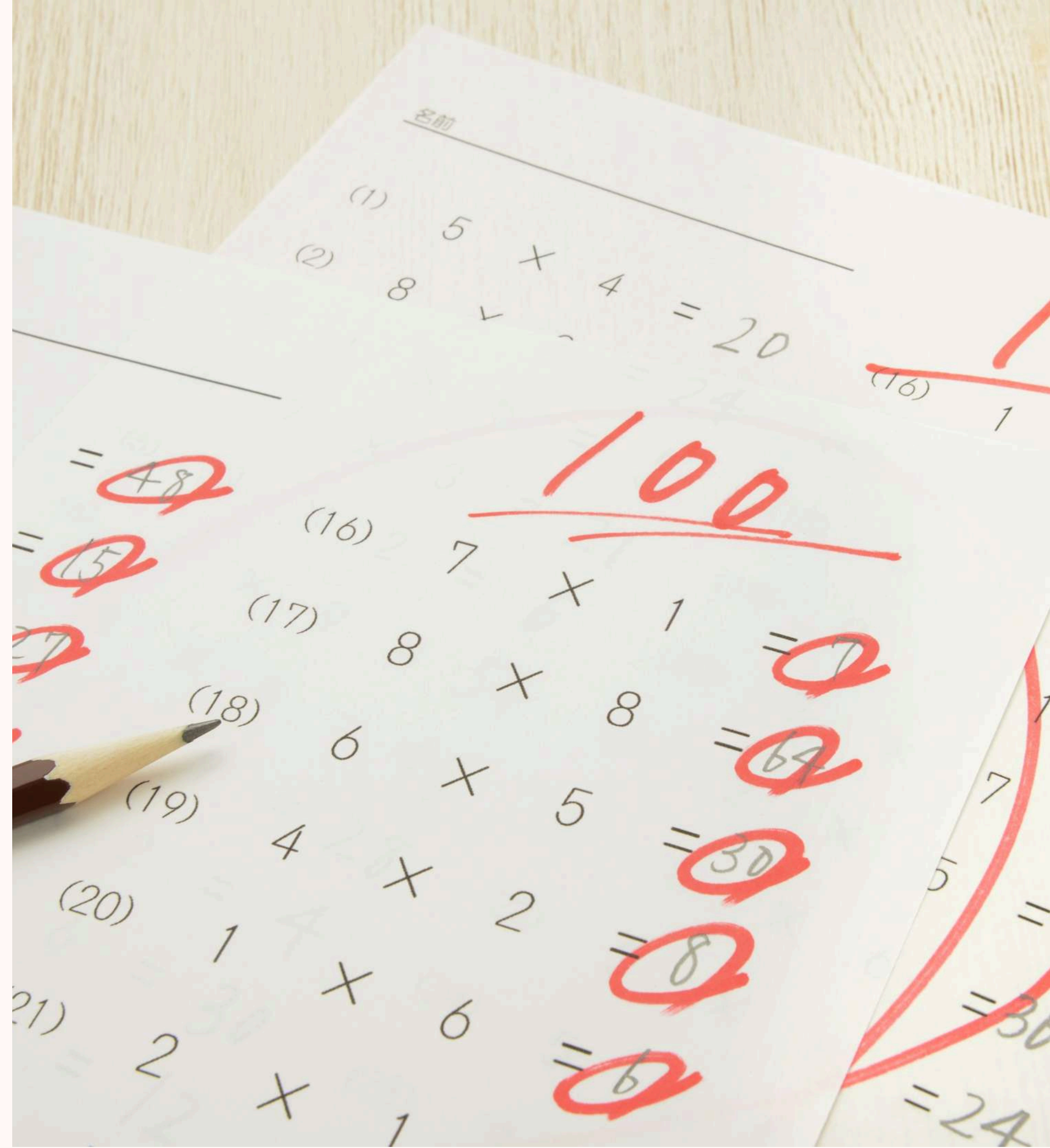
Calculating Z-scores

- **Z-score formula:** $Z = (X - \text{mean}) \div \text{standard deviation}$
- **X** represents your score in the dataset
- **Mean** is the average score of the group



Example of Z-score Calculation

To find the Z-score, use the formula: $Z = (X - \text{mean}) \div \text{standard deviation}$. For instance, with a mean of 50 and SD of 10, if your score is 65, the Z-score is 1.5.



Understanding Z-scores Interpretation

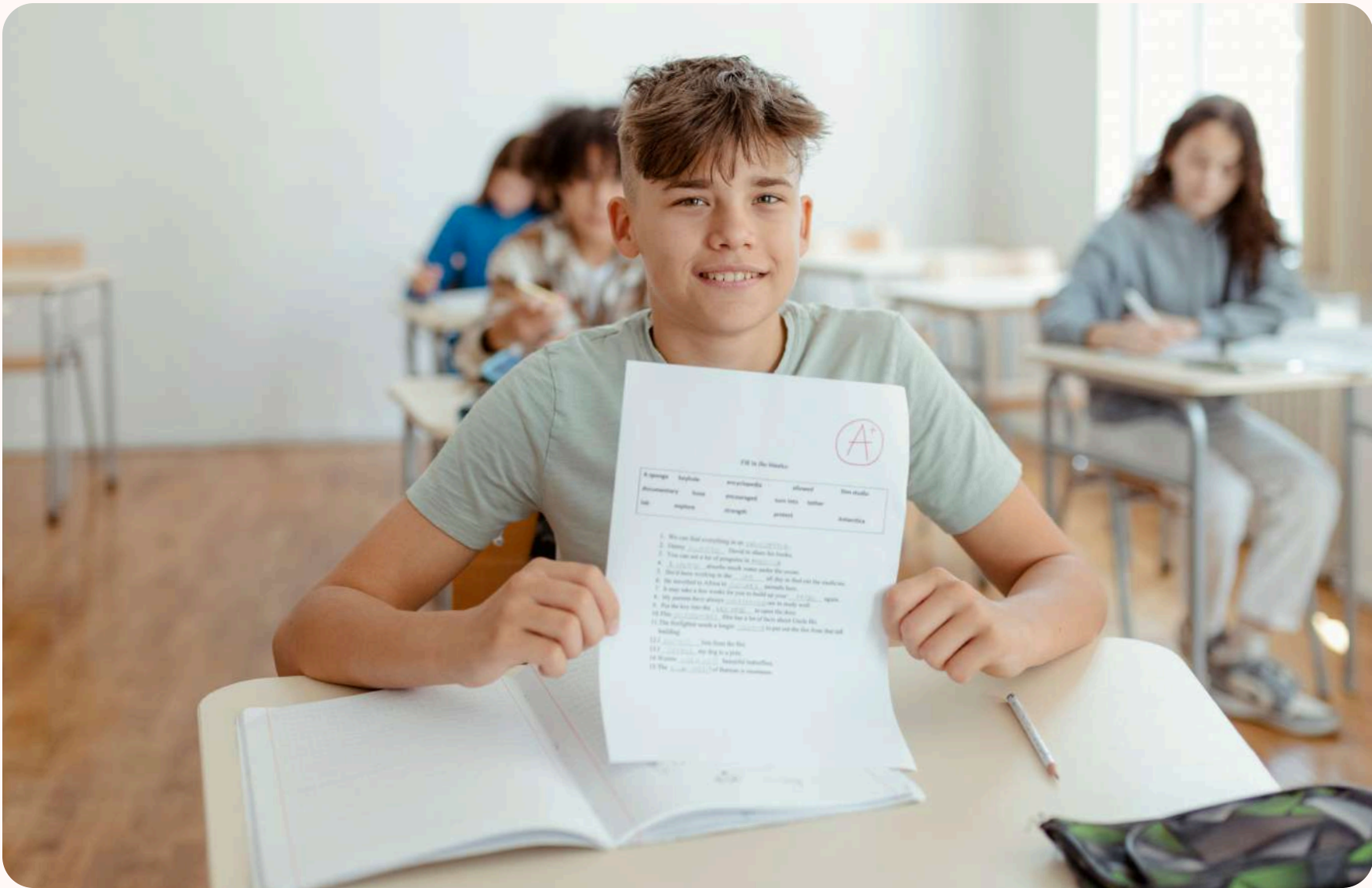
What Z-scores tell us about scores

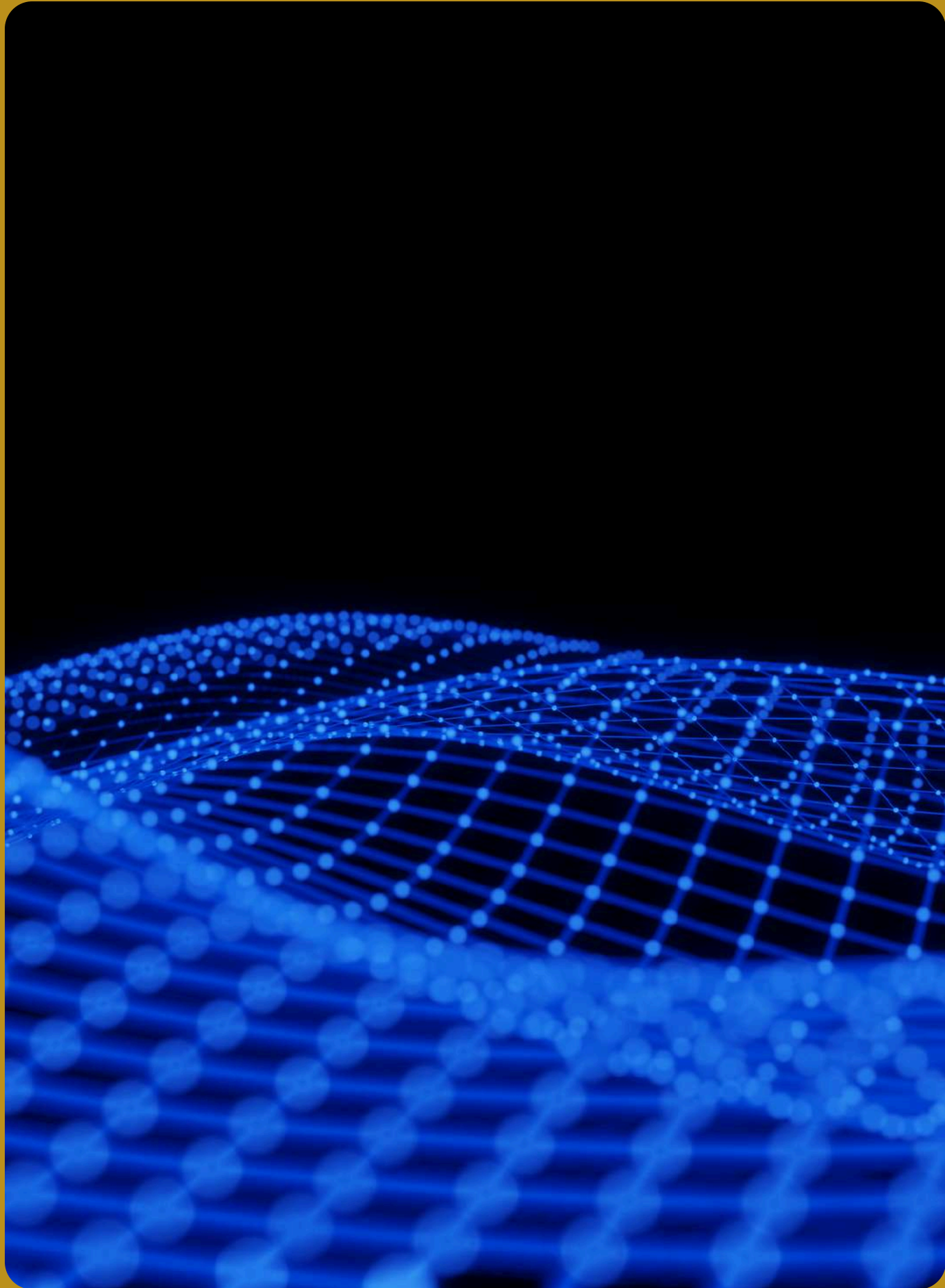
Z-scores help us determine a score's relationship to the average. A **positive Z-score** indicates above-average performance, while a **negative Z-score** shows below-average performance.



Comparing Z-scores

Understanding scores across different subjects





What is the Z-score if the score is 70?

1

2

3

4

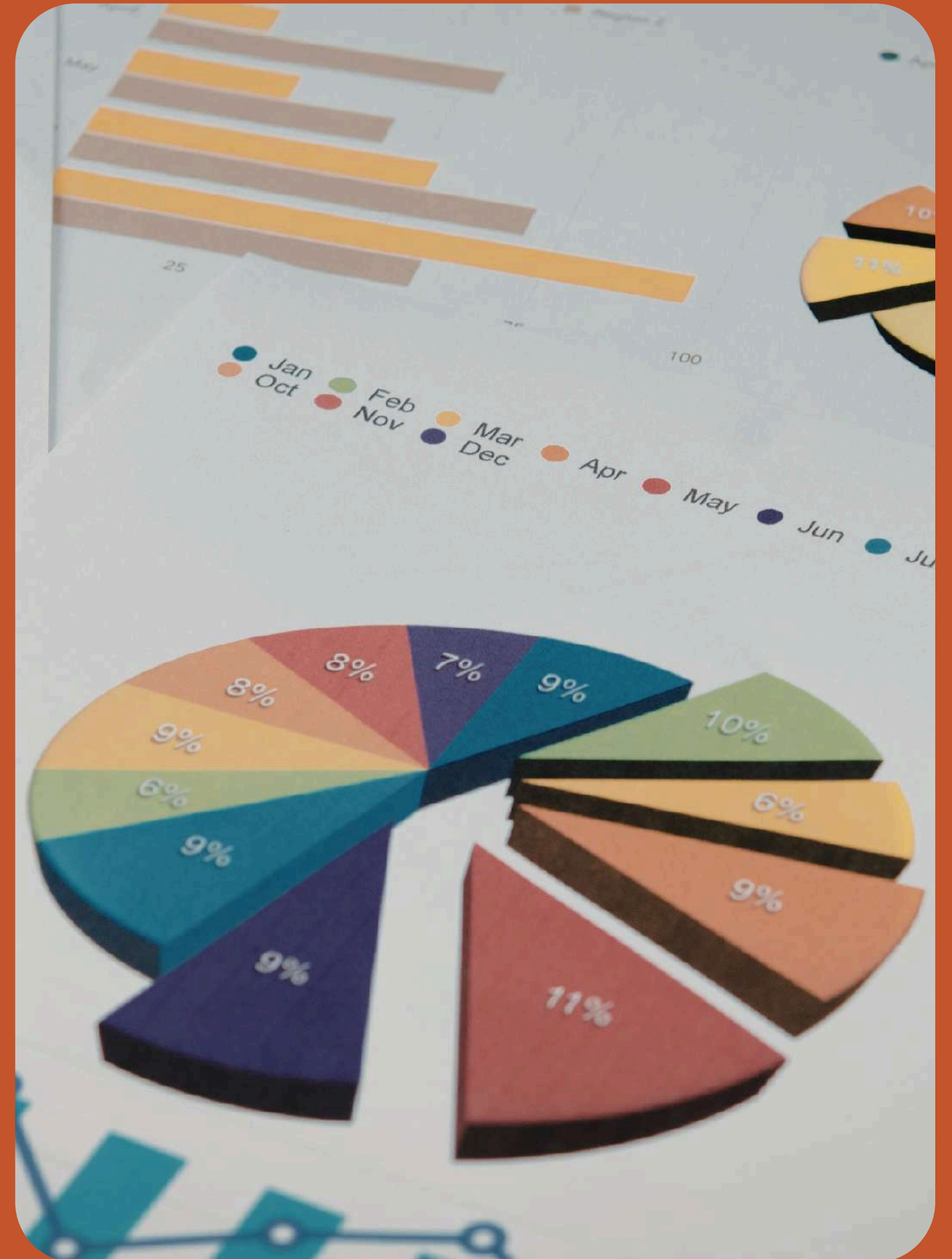
What does a Z-score of -1 indicate?

Below average

Above average

Average score

Significantly high



Recap of Key Points



01

Normal Distribution

It resembles a **bell curve** shape.

02

Z-scores

They indicate **how far** a score is from average.

03

Importance

Understanding this helps us **analyze data better**.

Real-Life Importance of Statistics

Understanding statistics helps you grasp **real-world applications** like sports scores and exam results, making math fun and relevant. Embrace these skills to enhance your decision-making and problem-solving in everyday life!





Thank You

We hope you enjoyed learning!